

Day 30: Capstone Project — Embedded C++ Application

Platform: nRF52840 | RTOS: Zephyr RTOS

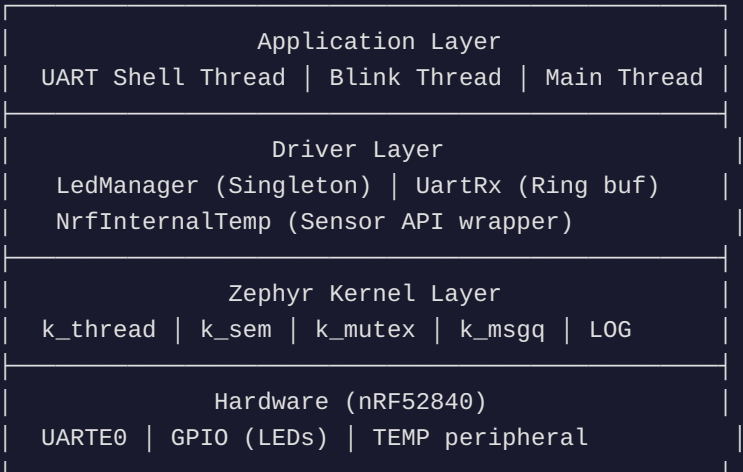
Learning Objectives

Project Overview

This capstone demonstrates everything from the 30-day plan in a single, complete Zephyr application running on nRF52840:

- **UART command shell** with AT-style command parsing
- **LED control** via commands with a background blink FSM
- **Temperature monitoring** using the nRF52840 internal temperature sensor
- **Multi-threaded design**: shell thread + blink thread + main heartbeat
- **OOP architecture**: Singleton `LedManager`, RAII locks, layered design

Architecture



Commands

Command	Description
<code>led on <0-3></code>	Turn LED N on
<code>led off <0-3></code>	Turn LED N off
<code>led blink <0-3></code>	Blink LED N at 1Hz
<code>temp</code>	Read internal die temperature
<code>status</code>	Show uptime and error count
<code>help</code>	List all commands

C++ Concepts Used

- **Day 3:** `uint8_t` , `constexpr` , `volatile`
- **Day 4:** `BIT()` , bit masking for LED pattern
- **Day 6:** Function pointers for command dispatch table
- **Day 7:** `LOG_MODULE_REGISTER` , `LOG_INF/ERR`
- **Day 13:** ISR + `k_sem_give` for UART RX
- **Day 15:** `LedManager` class, `PidController` class
- **Day 16:** RAIL `ScopedLock` for UART TX mutex
- **Day 20:** Singleton `LedManager::instance()`
- **Day 21:** `ITemperatureSensor` HAL + `NrfInternalTemp` driver
- **Day 22:** Return codes with `-EINVAL` , `__ASSERT`
- **Day 25:** LED blink state machine with `enum class`
- **Day 27:** `K_THREAD_DEFINE` for shell and blink threads
- **Day 28:** `K_SEM_DEFINE` , `K_MUTEX_DEFINE` for synchronization

Build & Flash

```
cd /home/eva/workspace/My-CPP_learning/src/day30
west build -b nrf52840dk/nrf52840 .
west flash
# Open serial terminal at 115200 baud:
```

```
picocom -b 115200 /dev/ttyACM0
# Then type: help
```

Extension Ideas

After completing the capstone, extend it:

1. Add BLE GATT characteristic that mirrors the UART commands
 2. Add ADC channel reading with `adc` command
 3. Add SPI sensor (BME280) with `sensor` command
 4. Add Flash storage for command history and configuration
 5. Add OTA firmware update support via Zephyr MCUboot
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Practice Tasks

1. Build and flash the capstone — test all 6 commands over serial
 2. Add an `adc` command that reads an ADC channel and reports voltage in mV
 3. Add a command history buffer (last 8 commands) with UP arrow recall (ANSI escape codes)
 4. Port the UART shell to BLE: the same command handler, but input via a BLE GATT characteristic
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Build & Run

```
cd /home/eva/workspace/My-CPP_learning/src/day30
west build -b nrf52840dk/nrf52840 .
west flash
# View output via RTT or UART serial (115200 baud)
west attach # RTT viewer (requires J-Link)
```

Resources

- [Zephyr Docs](#)
- [nRF52840 Product Specification](#)
- [Nordic DevZone](#)
- Source: [src/day30/main.cpp](#)